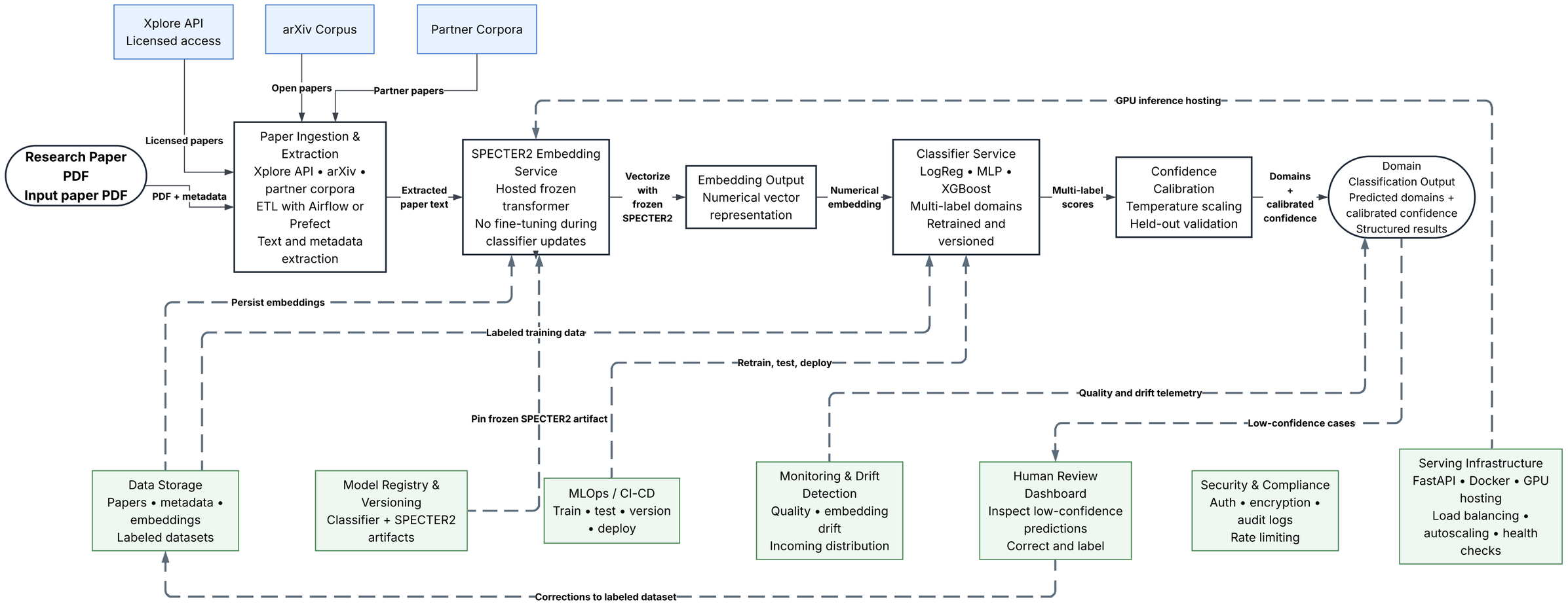

Xplore API
Licensed access
arXiv Corpus
Partner Corpora
Open papers
Partner papers
GPU inference hosting
Licensed papers
Research Paper PDF
Input paper PDF
PDF + metadata
Paper Ingestion & Extraction
Xplore API • arXiv • partner corpora
ETL with Airflow or Prefect
Text and metadata extraction
Extracted paper text
SPECTER2 Embedding Service
Hosted frozen transformer
No fine-tuning during classifier updates
Vectorize with frozen SPECTER2
Embedding Output
Numerical vector representation
Numerical embedding
Classifier Service
LogReg • MLP • XGBoost
Multi-label domains
Retrained and versioned
Multi-label scores
Confidence Calibration
Temperature scaling
Held-out validation
Domains + calibrated confidence
Domain Classification Output
Predicted domains + calibrated confidence
Structured results
Persist embeddings
Labeled training data
Retrain, test, deploy
Quality and drift telemetry
Low-confidence cases
Pin frozen SPECTER2 artifact
Data Storage
Papers • metadata • embeddings
Labeled datasets
Model Registry & Versioning
Classifier + SPECTER2 artifacts
MLOps / CI-CD
Train • test • version • deploy
Monitoring & Drift Detection
Quality • embedding drift
Incoming distribution
Human Review Dashboard
Inspect low-confidence predictions
Correct and label
Security & Compliance
Auth • encryption • audit logs
Rate limiting
Serving Infrastructure
FastAPI • Docker • GPU hosting
Load balancing • autoscaling • health checks
Corrections to labeled dataset